



TIME-BASED DATA

Applied Maths Shortcut Guide — By Thunderstudy

When values change over time (Daily, Yearly), we use patterns to predict future values. Focus on the trend!

⚡ 2-Second Trend Direction Trick

Look only at the **First** and **Last** values:

- **Last > First:** Upward Trend 
- **Last < First:** Downward Trend 
- **Last $\approx$ First:** Stable / No Trend

⚡ Missing Value & Forecasting Tricks

Missing Value (Smooth Pattern): Mid-term $\approx$ Average of neighbors.

Example: 20, ?, 40 $\rightarrow ? = (20+40)/2 = 30$.

Simple Forecast: **Last Value + Average Change.**

$$\text{Average Change} = (\text{Last} - \text{First}) / \text{Number of Gaps}$$

⚡ Trend Recognition: Linear vs Compound

- **Linear Trend:** If the Difference is constant (e.g., +4, +4, +4).

- **Compound/Growth:** If the Percentage (%) is constant (e.g., $100 \rightarrow 110 \rightarrow 121$ is +10% growth).



10-Second MCQ Strategy

1. Observe pattern (Increasing? Decreasing?).
2. Check difference once.
3. If differences are equal $\rightarrow$ Linear.
4. If % is same $\rightarrow$ Compound Growth.
5. Use **Average Change** for quick predictions without a table.

Question Type	Shortcut Rule
Trend Direction	Last – First
Next Value	Last + Avg Change
Missing Value	Average of neighbors
Same Difference	Linear Trend
Same % change	Compound Growth



MATCH PATTERN $\rightarrow$ WIN MARKS

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